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Exploratory Associations of Heart Rate and Running Velocity at Fixed Blood Lactate Concentrations with Sprint and Jump Performance in Male U19 Soccer Players

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Abstract

Background: Relationships between heart rate and running velocity at fixed blood lactate concentrations with short-duration performance remain uncertain in U19 soccer players. This exploratory cross-sectional study examined their associations with 10 m sprint time and countermovement jump (CMJ) height. **Methods:** Sixteen male players from a U19 development squad completed CMJ and 10 m sprint assessments followed by an incremental field-running test. The best result from two CMJ and two sprint trials was analyzed. Pearson correlations were reported with 95% confidence intervals, and the Benjamini–Hochberg procedure controlled the false discovery rate (FDR) across 24 tests. **Results:** Nominal positive correlations were observed between 10-m sprint time and heart rate at fixed blood lactate concentrations of 2.0–3.5 mmol·L^{-1} ($r = 0.503$–0.579; raw $p = 0.019$–0.047), indicating that higher heart rate was associated with longer sprint time. None remained statistically significant after FDR correction ($q \geq 0.207$). No CMJ or running-velocity association survived FDR correction. **Conclusions:** The data did not provide multiplicity-corrected evidence of associations between fixed-concentration test responses and sprint or jump performance. The estimates are exploratory and require confirmation in larger, prospectively powered samples.

Keywords: blood lactate; countermovement jump; field testing; heart rate; soccer; sprint performance; U19

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1. Introduction

Soccer is characterized by intermittent high-intensity activity in which performance depends on the integrated contribution of aerobic and anaerobic energy systems together with explosive performance capacity [1]. In recent years, the physical demands of the game have increased, particularly with respect to high-intensity running, repeated accelerations, and sprint-related actions [2]. At the same time, declines in physical output are often observed as match play progresses, highlighting the importance of both metabolic efficiency

and repeated high-intensity performance capacity [3]. Accordingly, training approaches integrating metabolic and performance-related components have been shown to improve sprinting and jumping outcomes [4].

Sprint and countermovement jump (CMJ) performance are widely used field-based indicators of explosive performance in soccer [5]. Consistent associations between these variables have been reported across different populations [6,7], supporting their use as complementary performance indicators in applied soccer settings.

Running velocity and heart rate at fixed blood lactate concentrations are commonly used to characterize endurance-related responses during incremental exercise [8]. Previous studies have examined how such physiological indices relate to sprint and jump performance, with mixed findings [9,10]. Some evidence supports their relevance to field performance [11], whereas other work indicates that aerobic indices alone cannot explain short-duration performance [12]. Differences in testing protocols, competitive level, and laboratory versus field assessment may contribute to these inconsistencies [13]. In applied soccer settings, physiological and performance-related assessments are used to characterize players' physical profiles [14,15]. Endocrine markers have also been examined in professional and amateur soccer players as complementary indicators of physiological status [16].

Morphological characteristics may provide descriptive context for performance testing. A mesomorphic profile has been associated with sprint and jump ability [17], while anthropometric and physical characteristics may differ across player populations [18–20]. Moreover, the physical and physiological demands of soccer match play vary according to playing position [21]. In the present study, somatotype was used only to characterize the study cohort and was not treated as an inferential performance predictor.

Although sprint, jump, and blood lactate test responses have been widely studied, less is known about the associations between running velocity and heart rate at fixed blood lactate concentrations and explosive performance measures within the same cohort of U19 soccer players. Sprint and jump tests primarily reflect explosive performance, whereas running velocity and heart rate at fixed blood lactate concentrations describe endurance-related responses [22–24]. Examining these measures in the same players may provide preliminary insight into their relationships and potential relevance for field-based monitoring.

Therefore, this exploratory study examined associations of running velocity and heart rate at fixed blood lactate concentrations with 10 m sprint time and CMJ height in male U19 soccer players under field-based conditions.

2. Materials and Methods

2.1. Study Design and Procedures

This cross-sectional, single-group study assessed physiological and field-based performance within one session on a natural grass field. After the standardized warm-up, tests were completed in the following order: CMJ, 10 m sprint, and the incremental blood lactate test. Participants received 5 min of passive recovery between the CMJ and sprint assessments and 10 min between the sprint assessment and the incremental test. Scheduling the exhaustive incremental test last minimized its potential influence on sprint and jump performance.

2.2. Participants

Sixteen male players from the U19 development squad of a professional club affiliated with the Turkish Football Federation Second League volunteered for the study (age: 18.56 ± 0.73 years; range: 17–19 years). All players competed in the national youth devel-

opment system and had at least five years of competitive soccer experience and three years of structured club-based training. Participant characteristics are presented in Table 1.

Table 1. Participant characteristics and somatotype components (n = 16).

Variable	Mean ± SD	Min–Max
Age (years)	18.56 ± 0.73	17.00–19.00
Height (cm)	176.49 ± 6.28	162.00–187.20
Body mass (kg)	71.49 ± 6.02	61.30–81.10
BMI ($kg \cdot m^{-2}$)	22.9 ± 1.5	20.1–25.9
Endomorphy	2.71 ± 0.57	1.99–3.84
Mesomorphy	4.30 ± 0.70	3.34–5.88
Ectomorphy	2.42 ± 0.59	1.10–3.20

Inclusion criteria required participants to be free from musculoskeletal injury during the preceding three months and to provide written informed consent. Exclusion criteria included any known cardiovascular, respiratory, or metabolic disorders; musculoskeletal injury within three months prior to testing; any clinical condition limiting maximal-effort performance; symptoms of acute illness or fatigue on the testing day; and the use of medications or ergogenic aids that could affect performance. Athletes who did not comply with the testing protocol or failed to complete all assessments were excluded from the analysis.

Participants were instructed to refrain from intense physical activity and to avoid caffeine intake for at least 48 h prior to testing.

The study protocol was approved by the Çankırı Karatekin University Health Sciences Ethics Committee (Meeting No. 24; 10 January 2026) before data collection. All procedures followed the Declaration of Helsinki. Written informed consent was obtained from all participants and, where applicable, from their legal guardians before participation.

2.3. Familiarization and Demonstration Session

To ensure protocol adherence and enhance measurement reliability, a structured familiarization session was conducted before the main testing session [24]. The session was performed under field conditions consistent with those used during testing. Participants were instructed on the procedures for the sprint, countermovement jump (CMJ), and incremental running test, including execution technique and performance requirements. Sub-maximal practice trials were completed to reduce technical variability and improve test consistency.

2.4. Warm-Up Protocol

Participants completed a standardized, progressive dynamic warm-up lasting approximately 10–15 min prior to testing. The protocol consisted of four sequential phases: general activation, movement preparation, neuromuscular activation, and sport-specific readiness. The warm-up included low-intensity running, dynamic mobility exercises targeting the lower extremities, and neuromuscular activation drills such as lunges, skipping, and high-knee movements. In the final phase, participants performed progressive accelerations and low-volume plyometric drills to prepare for sprint and jump testing. All warm-up procedures were supervised by the team's athletic performance coach to ensure consistent implementation across participants and standardized physiological preparation prior to performance assessments.

2.5. Anthropometric Measurements and Somatotype

Stature was measured using a Holtain stadiometer (Holtain Ltd., Crymych, UK; accuracy: 0.1 cm), and body mass was assessed with a Seca digital scale (Germany; accuracy: 0.1 kg). Skinfold thickness was measured at the triceps, subscapular, supraspinale, and medial calf sites, and the required girth and breadth measurements were obtained in accordance with International Society for the Advancement of Kinanthropometry protocols. These measurements were collected specifically to calculate Heath-Carter somatotype components; body-fat percentage and fat-free mass were not estimated and were not study outcomes. The same experienced investigator performed all measurements using standardized procedures.

Somatotype components (endomorphy, mesomorphy, and ectomorphy) were calculated using the Heath–Carter anthropometric method [25]. Somatotype assessment was used to characterize the morphological profile of the participants. For descriptive purposes, participants' somatotype distributions were visualized using a somatogram based on the Heath–Carter classification.

2.6. Incremental Running Test and Fixed Blood Lactate Concentrations

Following the standardized warm-up and the recovery periods described above, participants completed an incremental running protocol on a natural grass field. The test began at 8 km·h^{-1}, and running speed increased by 1 km·h^{-1} every three minutes until voluntary exhaustion or inability to maintain the required speed. At the end of each stage, a capillary blood sample was collected from the earlobe and analyzed using a portable lactate analyzer (Lactate Scout+, EKF Diagnostics, Barleben, Germany).

Running velocity and heart rate corresponding to blood lactate concentrations of 2.0, 2.5, 3.0, 3.5, 4.0, and 5.0 mmol·L^{-1} were estimated by linear interpolation between the two adjacent measured stages that bracketed each target concentration. Heart rate was recorded continuously using a Polar H10 sensor [14]. In this manuscript, vLT denotes the interpolated running velocity at a specified fixed blood lactate concentration rather than an individually determined physiological threshold.

2.7. Sprint Performance Test (10 m Sprint; MS10)

Sprint performance was assessed using a 10 m sprint test conducted under standardized field conditions. Sprint times were recorded with millisecond precision using a Witty photocell timing system (Microgate, Bolzano, Italy). Timing gates were positioned at the start and finish lines, and participants began each trial from a standardized standing start with the lead foot placed just behind the starting line.

Participants completed two maximal 10 m sprint trials separated by two minutes of passive recovery. Within-session reliability was calculated using a two-way mixed-effects, absolute-agreement, single-measure intraclass correlation coefficient [ICC(A,1)] and was 0.802. The fastest time was retained for analysis.

2.8. Countermovement Jump (CMJ)

CMJ height was measured using a Kistler force platform (Kistler Instruments, Winterthur, Switzerland), calibrated according to the manufacturer's instructions. Participants kept their hands on their hips and used a self-selected countermovement depth. Two maximal trials were completed with 60 s of passive recovery between attempts. Within-session reliability, calculated using a two-way mixed-effects, absolute-agreement, single-measure ICC [ICC(A,1)], was 0.905. The highest jump was retained for analysis.

2.9. Test Standardization and Measurement Reliability

All testing took place on a natural grass field familiar to the players between 09:00 and 12:00. Ambient temperature (20–24 °C) and relative humidity (40–50%) were monitored with a digital thermohygrometer (accuracy: ±0.5 °C and ±3% relative humidity). The same experienced researchers administered all assessments. Test order and inter-test recovery were identical for every participant.

Sprint and countermovement jump tests were selected because they are widely used field-based assessments with established practical relevance in athletic populations [5,22,24].

2.10. Statistical Analysis

Analyses were performed using SPSS version 26.0 (IBM Corp., Armonk, NY, USA) and independently verified against the raw dataset. Distributional assumptions were assessed using the Shapiro–Wilk test and Q-Q plots. Descriptive statistics are reported as mean ± standard deviation (SD) and range. Pearson product-moment correlations examined associations of 12 fixed-concentration indicators (six running velocities and six heart-rate values) with two outcomes (best 10 m sprint time and highest CMJ), producing 24 tests. Each correlation is reported with a two-sided raw p value and a 95% confidence interval calculated using Fisher's z transformation. To account for multiple comparisons, the Benjamini–Hochberg procedure was applied across all 24 correlation tests to control the false discovery rate (FDR). Adjusted q values are reported throughout the manuscript, with $q < 0.05$ considered statistically significant after correction. No formal a priori sample size calculation was performed because all eligible players in the available U19 squad were included. With $n = 16$ and two-sided $\alpha = 0.05$, the study had 80% power only for correlations of approximately $|r| \geq 0.63$; the analyses were therefore treated as exploratory. Within-session reliability for the two CMJ and two sprint trials was estimated using two-way mixed-effects, absolute-agreement, single-measure ICCs [ICC(A,1)].

3. Results

The results of the study are presented in terms of descriptive statistics and correlation analyses examining the relationships between physiological and performance variables.

The participants were male youth soccer players with a mean age of 18.56 ± 0.73 years, height of 176.49 ± 6.28 cm, body mass of 71.49 ± 6.02 kg, and BMI of 22.9 ± 1.5 kg·m^{-2} (Table 1). The mean somatotype component scores were 2.71 ± 0.57 for endomorphy, 4.30 ± 0.70 for mesomorphy, and 2.42 ± 0.59 for ectomorphy, with mesomorphy exhibiting the highest mean value.

Descriptive statistics for performance and physiological variables are shown in Table 2. Best 10 m sprint time was 1.695 ± 0.055 s, and the highest CMJ was 38.69 ± 5.49 cm. At a fixed blood lactate concentration of 3.0 mmol·L^{-1}, mean heart rate was 165.98 ± 11.17 beats·min^{-1} and interpolated running velocity was 12.43 ± 0.96 km·h^{-1}.

The correlations of heart rate and running velocity at fixed blood lactate concentrations with CMJ height and 10 m sprint time are presented in Table 3.

Table 2. Descriptive statistics of performance and physiological variables (n = 16).

Variable	Mean ± SD	Unit
10 m sprint time (best trial)	1.695 ± 0.055	s
CMJ height (highest trial)	38.69 ± 5.49	cm
HR at vLT2	151.89 ± 12.49	bpm
HR at vLT2.5	159.47 ± 11.65	bpm

Table 2. *Cont*.

Variable	Mean ± SD	Unit
HR at vLT3	165.98 ± 11.17	bpm
HR at vLT3.5	171.09 ± 10.35	bpm
HR at vLT4	174.95 ± 9.68	bpm
HR at vLT5	180.12 ± 8.09	bpm
Running velocity at vLT2	10.81 ± 1.32	$km \cdot h^{-1}$
Running velocity at vLT2.5	11.70 ± 1.12	$km \cdot h^{-1}$
Running velocity at vLT3	12.43 ± 0.96	$km \cdot h^{-1}$
Running velocity at vLT3.5	13.04 ± 0.87	$km \cdot h^{-1}$
Running velocity at vLT4	13.53 ± 0.80	$km \cdot h^{-1}$
Running velocity at vLT5	14.27 ± 0.82	$km \cdot h^{-1}$

The 10 m sprint value is the fastest of two trials, and CMJ height is the highest of two trials. HR = heart rate; vLT = interpolated running velocity at the indicated fixed blood lactate concentration.

Table 3. Pearson correlations of fixed blood lactate concentration indicators with countermovement jump height and 10 m sprint time (n = 16).

Outcome	Fixed-Concentration Indicator	r	95% CI	Raw *p*	FDR q
CMJ	RS vLT2	0.277	−0.254 to 0.679	0.300	0.577
	HR vLT2	−0.116	−0.578 to 0.403	0.670	0.797
	RS vLT2.5	0.294	−0.237 to 0.689	0.270	0.577
	HR vLT2.5	−0.092	−0.562 to 0.423	0.735	0.797
	RS vLT3	0.312	−0.217 to 0.700	0.239	0.573
	HR vLT3	−0.106	−0.572 to 0.411	0.697	0.797
	RS vLT3.5	0.384	−0.139 to 0.739	0.143	0.428
	HR vLT3.5	−0.079	−0.553 to 0.434	0.772	0.797
	RS vLT4	0.421	−0.094 to 0.759	0.104	0.357
	HR vLT4	−0.081	−0.555 to 0.432	0.764	0.797
	RS vLT5	0.460	−0.046 to 0.778	0.073	0.351
	HR vLT5	−0.136	−0.592 to 0.386	0.615	0.797
10 m sprint time	RS vLT2	0.230	−0.300 to 0.651	0.392	0.588
	HR vLT2	0.566	0.098 to 0.829	0.022	0.207
	RS vLT2.5	0.263	−0.268 to 0.671	0.325	0.577
	HR vLT2.5	0.579	0.117 to 0.835	0.019	0.207
	RS vLT3	0.257	−0.273 to 0.668	0.336	0.577
	HR vLT3	0.554	0.081 to 0.824	0.026	0.207
	RS vLT3.5	0.233	−0.297 to 0.653	0.386	0.588
	HR vLT3.5	0.503	0.010 to 0.800	0.047	0.281
	RS vLT4	0.193	−0.335 to 0.628	0.474	0.670
	HR vLT4	0.439	−0.073 to 0.768	0.089	0.356
	RS vLT5	0.070	−0.441 to 0.547	0.797	0.797
	HR vLT5	0.335	−0.192 to 0.713	0.204	0.544

CI = confidence interval; CMJ = countermovement jump; FDR = false discovery rate; HR = heart rate; RS = running velocity. Raw *p* values are two-sided. FDR q values were calculated with the Benjamini–Hochberg procedure across all 24 correlations. No association met $q < 0.05$.

Nominal positive correlations were observed between 10 m sprint time and heart rate at fixed blood lactate concentrations of 2.0 mmol·L^{-1} (r = 0.566, 95% CI [0.098, 0.829], raw $p = 0.022$), 2.5 mmol·L^{-1} (r = 0.579, 95% CI [0.117, 0.835], raw $p = 0.019$), 3.0 mmol·L^{-1} (r = 0.554, 95% CI [0.081, 0.824], raw $p = 0.026$), and 3.5 mmol·L^{-1} (r = 0.503, 95% CI [0.010, 0.800], raw $p = 0.047$). None remained significant after FDR correction (all q ≥ 0.207). Because sprint time was the outcome, positive coefficients indicate that higher heart rate was associated with slower sprint performance. Running velocity at the six fixed concentrations was not associated with sprint time after FDR correction. CMJ correlations ranged from r = −0.136 to 0.460. All 95% confidence intervals included zero, and no association with running velocity or heart rate was statistically significant before or after FDR correction.

4. Discussion

The main findings of the present study were that four positive correlations between 10 m sprint time and heart rate at fixed blood lactate concentrations of 2.0–3.5 mmol·L^{-1} met the conventional raw $p < 0.05$ threshold, whereas CMJ height was not significantly associated with the fixed-concentration variables. However, none of the 24 associations remained statistically significant after FDR correction. The findings therefore do not provide multiplicity-corrected evidence of an association between heart rate or running velocity at fixed blood lactate concentrations and sprint or jump performance in this sample. Rather, the nominal heart-rate pattern should be regarded as exploratory and hypothesis-generating. Values at neighboring concentrations were derived from the same incremental test and were not independent. Previous work has demonstrated the practical use of heart rate and running velocity at fixed lactate concentrations in team-sport players [11,14], but the present results do not establish that these indices predict short-distance performance.

The direction of the nominal correlations also requires careful interpretation. Because sprint time, rather than sprint velocity, was the performance outcome, the positive coefficients indicate that higher heart rate was associated with a longer, and therefore slower, 10 m sprint time. The wide confidence intervals indicate substantial uncertainty in both the magnitude and, for several coefficients, the direction of the associations. Previous research has described relationships among lactate-related running responses, heart rate, aerobic fitness, and repeated-sprint performance in soccer [13,14], while training near physiological thresholds has been associated with changes in heart rate and running performance [23]. These studies provide useful context but differ from the present cross-sectional analysis and cannot explain the observed coefficients. Autonomic regulation, running economy, energy preservation, and neuromuscular function were not directly measured and should not be presented as mechanisms for the current findings.

Sprint performance is a multifactorial outcome influenced by mechanical, neuromuscular, technical, and physiological characteristics rather than by aerobic or metabolic variables alone [9,10,22]. Altuğ et al. found that seasonal changes in eccentric hamstring performance in professional football players were reflected in impulse-based force–time measures despite no significant overall time effect on maximal eccentric force, underscoring the value of assessing multiple mechanical outputs when characterizing neuromuscular function [26]. Cardiovascular responses obtained during an incremental field test may contribute descriptive information about a player's broader physiological profile, but they should not be considered primary indicators of 10 m sprint ability. The absence of FDR-corrected associations means that the nominal heart-rate findings cannot support predictive or causal claims or guide individual monitoring without confirmation in larger samples. Future research should prespecify fewer primary associations and determine whether fixed-concentration responses add information beyond direct sprint testing.

CMJ height did not demonstrate statistically significant associations with heart rate or running velocity at the fixed blood lactate concentrations. This finding is compatible with evidence indicating that jumping performance is influenced primarily by strength, power, mechanical, and stretch-shortening-cycle characteristics, whereas endurance-oriented physiological indices may provide different information [9,10,22]. However, the present null findings should not be interpreted as evidence that these variables are independent. The sample was small, all confidence intervals included zero, and the positive running-velocity point estimates at some concentrations were too imprecise for firm interpretation. Previous studies have also examined anthropometric and morphological characteristics in relation to sprint and jump performance [17,18,20]. In the present study, however, somatotype was included only to describe the U19 cohort and was not analyzed as an inferential performance predictor; therefore, no conclusion regarding somatotype-performance associations is warranted. A strength of the study is the standardized assessment of physiological, sprint, and jump characteristics within the same cohort under ecologically relevant field conditions. The use of the best of two sprint trials and the highest of two CMJ trials, together with the observed within-session reliability, also supports the consistency of the selected performance values.

This study has several limitations. The sample comprised 16 players from one male U19 squad, and no a priori sample size calculation was performed. Under an unadjusted single-test framework, this sample size provided approximately 80% power only for correlations of $|r| \geq 0.63$; smaller but potentially relevant associations may therefore have been missed, and the resulting confidence intervals were wide. The analysis also included 24 related correlations, increasing the risk of false-positive findings. Although FDR correction was applied, none of the associations remained significant after correction. The cross-sectional design precludes causal or predictive inference, and the use of a single testing session limits the assessment of variability across days. CMJ and sprint testing preceded the exhaustive incremental test, and standardized recovery periods were used, but short-term within-session effects cannot be completely excluded. Running velocity and heart rate at the fixed blood lactate concentrations were estimated by interpolation and should not be interpreted as individually determined physiological thresholds. Training loads before testing were not quantified and may have contributed to between-player variability. Moreover, the field-based protocol enhanced ecological relevance but did not provide the physiological control or additional measurements available in laboratory studies. Finally, the inclusion of players from one male squad limits generalizability to female players, other age groups, and different competitive levels. Larger, prospectively powered and preferably longitudinal studies are needed to evaluate these exploratory patterns and to provide more precise effect estimates.

5. Conclusions

In this small exploratory sample, nominal positive correlations were observed between 10 m sprint time and heart rate at several fixed blood lactate concentrations. None remained statistically significant after FDR correction, and no corrected association was identified for CMJ height or running velocity.

These estimates are imprecise and should not be used to infer causal mechanisms or guide individual monitoring decisions. Larger, prospectively powered studies with prespecified primary analyses are required to determine whether incremental blood lactate test responses provide information beyond direct sprint and jump assessments in U19 soccer players.

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Data Availability Statement: The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request but are not publicly available due to ethical and privacy considerations.

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